## Similar questions



Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

**1** In this question $k$ is a non-zero constant.

**(a)** Find and simplify the first three terms, in ascending powers of $x$, of the expansion of $(3+kx)^5$. **[3]**

**(b)** Given that the coefficient of $x^2$ is three times the coefficient of $x$, find the value of $k$. **[3]**

**2**

**(a)** Find the first four terms, in ascending powers of $x$, of the expansion of $(1+3x)^5$. **[3]**

**(b)** Hence find the coefficient of $x^3$ in the expansion of $(2-x)(1+3x)^5$. **[3]**

**Question 1**

**(a)** $3^5 + \binom{5}{1}3^4kx + \binom{5}{2}3^3k^2x^2 = 243 + 405kx + 270k^2x^2$

**(b)** $270k^2 = 3 \times 405k \Rightarrow 270k = 1215 \Rightarrow k = \dfrac{9}{2}$

**Question 2**

**(a)** $1 + 5(3x) + 10(3x)^2 + 10(3x)^3 = 1 + 15x + 90x^2 + 270x^3$

**(b)** Coefficient of $x^3$: $2 \times 270 + (-1) \times 90 = 540 - 90 = 450$